

THERMODYNAMICS

Formula • Derivation • Conditions • Master Atlas

Complete derivation companion for JEE Main & JEE Advanced (Physical Chemistry)

Sign convention used throughout: w is work done *on* the system (so compression $w > 0$, expansion $w < 0$); heat entering the system has $q > 0$. First law: $\Delta U = q + w$. All boxed results are final, exam-ready formulas; every box is preceded by its derivation and followed by the conditions under which it applies.

Contents

Part I – Foundations

1. System, surroundings, state functions
2. First law, PV-work
3. Enthalpy & heat capacities
4. Degrees of freedom, equipartition

Part II – Gas Processes

5. Isochoric / Isobaric
6. Isothermal (rev. & irrev.)
7. Adiabatic (rev. & irrev.)
8. Polytropic
9. Free & stoppered expansion

Part III – Entropy & 2nd Law

10. Entropy, Clausius inequality
11. Carnot engine
12. Entropy of processes
13. Third law

Part IV – Free Energy

14. Gibbs function, spontaneity
15. G vs P , T ; Gibbs–Helmholtz
16. ΔG° and equilibrium

Part V – Thermochemistry

17. Enthalpy of reaction, Hess's law
18. Standard enthalpies (formation, combustion, atomisation, bond, solution, hydration, neutralisation, ionisation, electron gain, lattice)
19. Kirchhoff's equation
20. Born–Haber cycle
21. Resonance energy

Part VI – Exam Tools

22. Master formula atlas
23. Keyword decision tree
24. Symbol & sign glossary

1 Foundations: System, State Functions & First Law

1.1 Basic definitions

System – part of universe under study. **Surroundings** – everything else. **Boundary** – interface between them (real boundary = wall).

- **Closed system:** exchanges only energy. **Open system:** exchanges energy & matter. **Isolated system:** exchanges neither.
- **Rigid wall:** $w_{PV} = 0$. **Non-rigid wall:** $w_{PV} \neq 0$. **Adiabatic wall:** $q = 0$. **Diathermic wall:** $q \neq 0$.
- **Extensive properties** (depend on amount): mass, V , U , H , S , G , heat capacity.
- **Intensive properties** (independent of amount): P , T , density, molar heat capacity, viscosity, emf.
- **State function:** depends only on initial/final state (U, H, S, G, P, V, T). **Path function:** depends on the path taken (q, w) – also called indefinite quantities.

Equation of state

Ideal gas: $PV = nRT$. Van der Waals (real gas): $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$.

1.2 First law of thermodynamics

Derivation. Let a system go from state A to B . If it absorbs heat q and work w is done *on* it, energy conservation gives the final internal energy as initial energy plus both transfers:

$$U_B = U_A + q + w$$

Subtracting U_A from both sides, and noting the difference depends only on end states (so U is a state function even though q, w are path dependent):

Formula 1.1 First Law of Thermodynamics

$$\Delta U = q + w$$

Conditions

Valid for any process (closed system). For a cyclic process $\oint dU = 0 \Rightarrow \sum q_i = -\sum w_i$ over any set of paths between two states.

For an infinitesimal change involving only PV work:

Formula 1.2 Differential First Law

$$dU = \delta q - P dV$$

1.3 PV-work

Derivation. Consider a frictionless piston enclosing gas of cross-section A . An external force $F = P_{ext}A$ pushes the piston a distance dx , doing work $\delta w = F dx = P_{ext}A dx$ on the system. If $V = (\ell - x)A$, then $dV = -A dx$, so $A dx = -dV$. Hence:

$$\delta w = P_{ext}A dx = -P_{ext} dV$$

Formula 1.3 PV-Work (on the system)

$$\delta w = -P_{ext} dV \quad \Rightarrow \quad w = -\int_{V_1}^{V_2} P_{ext} dV$$

Conditions

Sign check: **expansion** $dV > 0 \Rightarrow w < 0$ (system does work, loses energy). **Compression** $dV < 0 \Rightarrow w > 0$ (energy added to system).

Common Trap

Older textbooks (pre-IUPAC) define w as work done *by* the system, flipping every sign below. This book uses the IUPAC convention: w = work done *on* system.

For an irreversible process against constant P_{ext} :

Formula 1.4 Irreversible Work

$$w_{irr} = -P_{ext} \Delta V$$

For a **stoppered (partial) expansion** where gas expands against constant P_{ext} but is stopped before reaching mechanical equilibrium (so $P_{ext} \neq P_{final}$):

Formula 1.5 Stoppered Expansion Work

$$w = -P_{ext} \left(\frac{nRT_f}{P_f} - \frac{nRT_i}{P_i} \right)$$

1.4 Enthalpy

Derivation. Define enthalpy to combine U with the PV term that naturally appears in constant-pressure chemistry:

$$H = U + PV \quad \Rightarrow \quad dH = dU + P dV + V dP$$

At constant pressure ($dP = 0$), and using $\delta q_p = dU + P dV$ (first law at const. P):

$$\delta q_p = dH \quad \Rightarrow \quad q_p = \Delta H$$

Formula 1.6 Enthalpy

$$H = U + PV$$

$$\Delta H = \Delta U + \Delta(PV)$$

$$q_p = \Delta H$$

For processes with only PV -work:

Formula 1.7 Heat at Constant V and P

$$q_V = \Delta U$$

$$q_P = \Delta H$$

1.5 Heat capacities and the Mayer relation

Derivation. Heat capacity $C = \delta q/dT$. At constant volume, $dU = \delta q_V$ (since $w = 0$), so

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V$$

Derivation. At constant pressure, $\delta q_P = dH$, so

$$C_P = \left(\frac{\partial H}{\partial T} \right)_P$$

Formula 1.8 Heat Capacities

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V$$

$$C_P = \left(\frac{\partial H}{\partial T} \right)_P$$

Derivation. For 1 mole of ideal gas, $H = U + RT$. Differentiating w.r.t. T : $C_P = C_V + R$.

Formula 1.9 Mayer's Relation

$$C_P - C_V = R$$

Conditions

Valid only for an ideal gas (1 mole). Since U, H are functions of T only for an ideal gas, subscripts can be dropped: $dU = nC_V dT$, $dH = nC_P dT$ for *any* process of an ideal gas. Hence $\Delta U = \Delta H = 0$ for an isothermal process of an ideal gas.

1.6 Degrees of freedom & equipartition

Derivation. Law of equipartition: each translational/rotational degree of freedom contributes $\frac{1}{2}RT$ per mole to thermal energy; each active vibrational degree contributes RT (since vibration has both KE and PE terms). If a molecule has f_T translational, f_R rotational and f_V vibrational degrees:

$$U = \left(\frac{f_T + f_R}{2} \right) nRT + f_V nRT$$

Differentiate w.r.t. T at constant V , then apply Mayer's relation:

Formula 1.10 Equipartition Internal Energy & Heat Capacities

$$U = \frac{f}{2} nRT$$

$$C_V = \left(\frac{f_T + f_R}{2} + f_V \right) R$$

$$C_P = C_V + R$$

$$\gamma = \frac{C_P}{C_V}$$

Counting rules

$f_T = 3$ always. Linear molecule: $f_R = 2$, $f_V = 3N - 5$. Non-linear molecule: $f_R = 3$, $f_V = 3N - 6$ (N = number of atoms). Monatomic: $f_R = f_V = 0$.

Molecule	f_T	f_R	f_V	$C_V(\text{excl. vib})$	$C_P(\text{excl.})$	$\gamma(\text{excl.})$	$C_V(\text{incl.})$	$C_P(\text{incl.})$	$\gamma(\text{incl.})$
Monatomic (He)	3	0	0	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{3}$	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{3}$
Diatomic (N_2)	3	2	1	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5}$	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5}$
Triatomic linear (CO_2)	3	2	4	$\frac{7}{2}R$	$\frac{9}{2}R$	$\frac{9}{7}$	$\frac{7}{2}R$	$\frac{9}{2}R$	$\frac{9}{7}$
Triatomic non-linear (H_2O)	3	3	3	$3R$	$4R$	$\frac{4}{3}$	$6R$	$7R$	$\frac{7}{6}$

Common Trap

Vibrational modes are active only at high temperature; the experimentally observed C_V for diatomics near room temperature ($\approx \frac{5}{2}R$) matches the "excl. vibration" column, not the full equipartition value. Always check whether the question says "vibration neglected" or "all modes active."

2 Gas Processes

2.1 Isochoric, isobaric, cyclic processes

Definitions

Isochoric: $V = \text{const}$, $dV = 0 \Rightarrow w = 0$, $q_V = \Delta U = nC_V\Delta T$. For ideal gas: $P/T = \text{const}$ (Gay-Lussac's law).

Isobaric: $P = \text{const}$, $dP = 0 \Rightarrow w = -P\Delta V = -nR\Delta T$, $q_P = \Delta H = nC_P\Delta T$. For ideal gas: $V/T = \text{const}$.

Cyclic: system returns to initial state $\Rightarrow \Delta U = 0$, $\Delta H = 0$, $\Delta S = 0$ for *every* state function; hence $q = -w$ over the whole cycle.

Formula 2.1 Isochoric / Isobaric heat and work

$$w_V = 0, \quad q_V = \Delta U = nC_V\Delta T$$

$$w_P = -nR\Delta T, \quad q_P = \Delta H = nC_P\Delta T$$

2.2 Isothermal processes

2.2.1 Reversible isothermal work

Derivation. For an ideal gas at constant T , $PV = nRT \Rightarrow P = nRT/V$. In a reversible path $P_{ext} = P_{int}$ at every stage, so

$$w_{rev} = - \int_{V_1}^{V_2} P dV = -nRT \int_{V_1}^{V_2} \frac{dV}{V} = -nRT \ln\left(\frac{V_2}{V_1}\right)$$

Using $\ln x = 2.303 \log_{10} x$:

Formula 2.2 Reversible Isothermal Work

$$w_{rev} = -nRT \ln\left(\frac{V_2}{V_1}\right) = -2.303 nRT \log\left(\frac{V_2}{V_1}\right) = -nRT \ln\left(\frac{P_1}{P_2}\right)$$

Conditions

Ideal gas, T constant, reversible path. This is the *maximum* magnitude of work obtainable in expansion (or minimum required for compression) between the two states.

Derivation. Since U of an ideal gas depends only on T , and $T_2 = T_1$ here, $\Delta U = nC_V\Delta T = 0$. The first law then gives $q = -w$:

Formula 2.3 Isothermal ΔU , q

$$\Delta U = 0$$

$$q = -w$$

2.2.2 Entropy of isothermal expansion

Derivation. For a reversible process $dS = \delta q_{rev}/T$. Since T is constant and $q_{rev} = -w_{rev}$:

Formula 2.4 Isothermal Entropy Change

$$\Delta S = nR \ln\left(\frac{V_2}{V_1}\right) = nR \ln\left(\frac{P_1}{P_2}\right)$$

2.2.3 Irreversible isothermal expansion

Formula 2.5 Irreversible Isothermal Work

$$w_{irr} = -P_{ext} \Delta V$$

Key Points

Comparing magnitudes at same P_{ext} vs. reversible path (same initial & final states): $|w_{rev,expansion}| > |w_{irr,expansion}|$, while $|w_{rev,compression}| < |w_{irr,compression}|$. As the number of intermediate steps in a multi-step irreversible process $\rightarrow \infty$, $w_{irr} \rightarrow w_{rev}$.

2.2.4 Free expansion

When a gas expands against $P_{ext} = 0$ (into vacuum):

Formula 2.6 Free Expansion

$$w = 0, \quad q = 0, \quad \Delta U = 0, \quad \Delta T = 0 \text{ (ideal gas)}$$

Conditions

Free expansion of an ideal gas is simultaneously isothermal, adiabatic, and irreversible. For real gases, cooling/heating on free expansion depends on whether T is above or below the inversion temperature.

2.3 Adiabatic processes**2.3.1 Reversible adiabatic – full derivation**

Derivation. Adiabatic means $\delta q = 0$. First law: $dU = \delta w = -P dV$ (reversible). For n moles ideal gas $dU = nC_V dT$, so

$$nC_V dT = -P dV$$

Using $P = nRT/V$:

$$C_V \frac{dT}{T} = -R \frac{dV}{V} \Rightarrow C_V \frac{dT}{T} + R \frac{dV}{V} = 0$$

Integrate between states 1 and 2:

$$C_V \ln \frac{T_2}{T_1} + R \ln \frac{V_2}{V_1} = 0$$

Using $C_P - C_V = R$ and $\gamma = C_P/C_V$, so $R/C_V = \gamma - 1$:

$$\ln \frac{T_2}{T_1} = -(\gamma - 1) \ln \frac{V_2}{V_1} \Rightarrow \frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{\gamma-1}$$

Formula 2.7 Reversible Adiabatic Relations (Poisson's Equations)

$$TV^{\gamma-1} = \text{constant}$$

$$PV^\gamma = \text{constant}$$

$$T^\gamma P^{1-\gamma} = \text{constant}$$

Conditions

$q = 0$, reversible, ideal gas, γ constant (temperature-independent C_V). Choose the form matching the given variables: P, V known $\rightarrow PV^\gamma$; T, V known $\rightarrow TV^{\gamma-1}$; T, P known $\rightarrow T^\gamma P^{1-\gamma}$.

2.3.2 Adiabatic work

Derivation. Since $q = 0$, $\Delta U = w$. For an ideal gas $\Delta U = nC_V(T_2 - T_1)$, and $C_V = R/(\gamma - 1)$:

Formula 2.8 Reversible Adiabatic Work

$$w = \Delta U = nC_V(T_2 - T_1) = \frac{nR(T_2 - T_1)}{\gamma - 1} = \frac{P_2V_2 - P_1V_1}{\gamma - 1}$$

Conditions

Sign check: expansion ($T_2 < T_1$) $\Rightarrow w < 0$; compression ($T_2 > T_1$) $\Rightarrow w > 0$.

2.3.3 Irreversible adiabatic process

Derivation. Suppose P_{ext} is changed suddenly and the gas equilibrates. Then $\Delta U = w = -P_{ext}(V_f - V_i)$, and $\Delta U = nC_V(T_f - T_i)$. Equating and solving:

$$nC_V(T_f - T_i) = -P_{ext} \left(\frac{nRT_f}{P_f} - \frac{nRT_i}{P_i} \right)$$

Formula 2.9 Irreversible Adiabatic Work

$$w = \Delta U = nC_V(T_2 - T_1) = \frac{P_f V_f - P_i V_i}{\gamma - 1}$$

Common Trap

$PV^\gamma = \text{const}$ applies *only* to the reversible adiabatic path – never use it for an irreversible (sudden) adiabatic change. Solve T_f from the energy balance above instead.

Isothermal vs. adiabatic

Starting from the same state and expanding to the same final pressure: $|w_{rev, iso}| > |w_{rev, adia}|$ (heat flows in during isothermal expansion, cushioning the fall in U). On the P - V diagram, since $dP/dV|_{iso} = -P/V$ and $dP/dV|_{adia} = -\gamma P/V$ with $\gamma > 1$, the **adiabatic curve is always steeper** than the isothermal curve through the same point.

2.4 Polytropic process

Derivation. A polytropic path is defined by $PV^n = K$ (constant), where n is any real number, so $P = KV^{-n}$. Work on the system:

$$w = - \int_{V_1}^{V_2} KV^{-n} dV = -K \left[\frac{V^{1-n}}{1-n} \right]_{V_1}^{V_2} \quad (n \neq 1)$$

Since $K = P_1 V_1^n = P_2 V_2^n$, this simplifies to the endpoint form:

Formula 2.10 Polytropic Work

$$PV^n = \text{constant}$$

$$w = \frac{P_2 V_2 - P_1 V_1}{n - 1}$$

Special cases of n

$n = 0 \Rightarrow$ isobaric $n = 1 \Rightarrow$ isothermal (use $w = -nRT \ln(V_2/V_1)$ instead; the $n = 1$ case must be handled separately since $1/(n-1)$ diverges) $n = \gamma \Rightarrow$ adiabatic $n \rightarrow \infty \Rightarrow$ isochoric.

Derivation. Heat capacity for a polytropic process. From $\delta q = dU + P dV = nC_V dT + \frac{nRT}{V} dV$, so $C = nC_V + \frac{nRT}{V} \frac{dV}{dT}$. Differentiating $PV^n = k$ using $P = nRT/V$ gives $\frac{dV}{dT} = -\frac{V}{T} \cdot \frac{1}{n-1}$. Substituting:

Formula 2.11 Polytropic Heat Capacity

$$C = C_V + \frac{R}{1-n} \quad (\text{per mole})$$

3 Entropy & the Second Law

3.1 Second law statements

Equivalent statements

Kelvin: No cyclic process can convert heat completely into work with no other change. **Clausius:** Heat cannot flow spontaneously from a colder to a hotter body without external work. **Entropy statement:** The entropy of the universe (system + surroundings) increases in every spontaneous (irreversible) process.

3.2 Entropy – fundamental definition

Derivation. Entropy is defined via the reversible heat exchanged:

Formula 3.1 Entropy Definition

$$dS = \frac{\delta q_{rev}}{T} \quad \Delta S = \int_1^2 \frac{\delta q_{rev}}{T}$$

Conditions

S is a *state function* because $\oint \frac{\delta q_{rev}}{T} = 0$ for any reversible cycle (proved via the Carnot cycle below, then generalised).

3.2.1 Ideal-gas entropy

Derivation. From first law, $\delta q_{rev} = dU + P dV$. Divide by T ; for ideal gas $dU = nC_V dT$ and $P/T = nR/V$:

$$dS = \frac{nC_V dT}{T} + \frac{nR dV}{V}$$

Integrating:

Formula 3.2 Ideal-Gas Entropy Change

$$\Delta S = nC_V \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1} = nC_P \ln \frac{T_2}{T_1} + nR \ln \frac{P_1}{P_2}$$

Special cases

Isothermal: $\Delta S = nR \ln(V_2/V_1)$. Isochoric: $\Delta S = nC_V \ln(T_2/T_1)$. Isobaric: $\Delta S = nC_P \ln(T_2/T_1)$.

3.3 Carnot engine & efficiency

Derivation. A reversible Carnot cycle: (i) isothermal expansion at T_H from $(P_1, V_1) \rightarrow (P_2, V_2)$; (ii) reversible adiabatic expansion $T_H \rightarrow T_C$; (iii) isothermal compression at T_C ; (iv) reversible adiabatic compression back to start. Net work on the engine over one cycle equals $-(q_H + q_C)$. For the two isothermal legs,

$$q_H = nRT_H \ln \frac{V_2}{V_1}, \quad q_C = nRT_C \ln \frac{V_4}{V_3}$$

For the two adiabatic legs, $T_H V_2^{\gamma-1} = T_C V_3^{\gamma-1}$ and $T_C V_4^{\gamma-1} = T_H V_1^{\gamma-1}$, which on division gives $V_2/V_1 = V_3/V_4$. Substituting into the efficiency definition $\eta = -w_{net}/q_H = (q_H + q_C)/q_H$:

Formula 3.3 Carnot Efficiency

$$\eta = \frac{q_H + q_C}{q_H} = \frac{T_H - T_C}{T_H} = 1 - \frac{T_C}{T_H}$$

Conditions

T_H, T_C in kelvin; $T_H > T_C$; engine operates reversibly. Efficiency depends *only* on the reservoir temperatures, not on the working substance. This is the maximum possible efficiency between two reservoirs.

Derivation. For the Carnot cycle, $q_H/T_H + q_C/T_C = 0$, i.e., $\sum q_{rev}/T = 0$ for the cycle. Since this holds for *any* reversible cycle (each can be built from infinitesimal Carnot cycles):

Formula 3.4 Cyclic Integral – proof that S is a state function

$$\oint \frac{\delta q_{rev}}{T} = 0$$

3.4 Clausius inequality

Derivation. If any step of the Carnot cycle is made irreversible, the efficiency *decreases* below the reversible value $1 - T_C/T_H$, so $q_H/T_H + q_C/T_C < 0$. Generalising to any irreversible cyclic path:

Formula 3.5 Clausius Inequality

$$\oint \frac{\delta q}{T} \leq 0$$

(equality holds only for a reversible cycle).

3.5 Entropy change of an isolated system

Derivation. Take a system irreversibly from $A \rightarrow B$, then return reversibly $B \rightarrow A$. By Clausius inequality applied to the full (irreversible) cycle: $\sum_{A \rightarrow B} \frac{q_{irr}}{T} + \sum_{B \rightarrow A} \frac{q_{rev}}{T} \leq 0$. Since the reverse reversible path gives $\sum_{B \rightarrow A} \frac{q_{rev}}{T} = -\Delta S_{system}(A \rightarrow B)$, this rearranges to $\left(\frac{\delta q}{T}\right)_{A \rightarrow B} \leq dS_{system}$, i.e., $dS_{system} - \left(\frac{\delta q}{T}\right)_{A \rightarrow B} \geq 0$. Since $-\delta q/T$ is exactly $dS_{surroundings}$ for a reversible heat exchange with surroundings:

Formula 3.6 Entropy Increase Principle

$$dS_{system} + dS_{surroundings} \geq 0$$

$$\Delta S_{total, isolated} > 0 \text{ (spontaneous)}$$

Conditions

$dS_{surr} = -q_{actual}/T$ always (whether the actual process is reversible or not). Equality ($= 0$) holds only in the reversible limit; strict inequality (> 0) signals a spontaneous, irreversible process. At equilibrium in an isolated system, S_{total} is maximum.

3.6 Entropy change in phase transitions

Derivation. At the transition temperature T_{tr} , phase change is reversible at constant T, P , so $\Delta S_{total} = 0 \Rightarrow \Delta S_{system} = -\Delta S_{surr} = \Delta H_{tr}/T_{tr}$.

Formula 3.7 Entropy of Fusion / Vaporisation

$$\Delta S_{fusion} = \frac{\Delta H_{fusion}}{T_f}$$

$$\Delta S_{vap} = \frac{\Delta H_{vap}}{T_b}$$

Conditions

T_f = melting point, T_b = (normal) boiling point at 1 atm, both in kelvin. Valid because phase transitions at constant T, P are reversible processes.

3.7 Entropy change in a chemical reaction

Formula 3.8 Reaction Entropy

For $aA + bB \rightarrow cC + dD$:

$$\Delta_r S^\circ = \sum \nu S_{products}^\circ - \sum \nu S_{reactants}^\circ = (cS_C + dS_D) - (aS_A + bS_B)$$

Key Points

ΔS tends to be large and *positive* when Δn_g (moles of gas) increases, since $S_{gas} \gg S_{liquid} > S_{solid}$ typically.

3.8 Third law of thermodynamics**Formula 3.9 Third Law**

$S(T = 0) = 0$ for every perfectly crystalline element/compound

Derivation. This allows computation of *absolute* entropy from heat-capacity data:

$$S(T) = S(0) + \int_0^T \frac{C_p}{T} dT \quad (\text{plus } \Delta H_{tr}/T_{tr} \text{ for any phase change crossed})$$

Conditions

Experimentally $C_p \rightarrow 0$ as $T \rightarrow 0$ (as T^3 for non-metals; crossing over to $\propto T$ for metals at very low T due to conduction electrons), which keeps the integral finite despite the $1/T$ factor.

3.9 Criteria of spontaneity via entropy**Formula 3.10**

$\Delta S_{system} > 0$ alone is *not* sufficient

$\Delta S_{system} + \Delta S_{surroundings} > 0$ is the universal criterion

4 Gibbs Free Energy & Spontaneity

4.1 Definition and the spontaneity criterion

Derivation. Entropy is a universal criterion, but requires tracking surroundings. Define the Gibbs function to give a system-only criterion at constant T, P :

$$G = H - TS$$

From the second law, $\delta q \leq T dS$ (irreversible/reversible). Using $\delta q = dU + P dV$:

$$dU + P dV - T dS \leq 0$$

Since $d(H - TS) = dU + P dV + V dP - T dS - S dT$, at constant T, P ($dP = dT = 0$) this reduces exactly to the left side above:

$$d(H - TS)_{T,P} \leq 0$$

Formula 4.1 Gibbs Free Energy & Spontaneity

$$G = H - TS$$

$$dG_{T,P} \leq 0$$

Criteria at constant T, P

$\Delta G < 0$: spontaneous. $\Delta G = 0$: equilibrium. $\Delta G > 0$: non-spontaneous (reverse process spontaneous).

4.2 G and total entropy

Derivation. At constant T, P : $\Delta G = \Delta H - T\Delta S = q_P - T\Delta S$. Dividing by $-T$ and using $\Delta S_{surr} = -q_P/T$:

$$-\frac{\Delta G}{T} = -\frac{q_P}{T} + \Delta S = \Delta S_{surr} + \Delta S_{system}$$

Formula 4.2 G – Total Entropy Link

$$-\Delta G = T \Delta S_{total}$$

$$\Delta G = \Delta H - T\Delta S$$

4.3 Gibbs function and non-expansion work

Derivation. For a closed system doing non- PV work w_{non-PV} in addition to PV -work: $dU = \delta q - P dV + \delta w_{non-PV}$ (careful sign bookkeeping as in text). For a reversible change at constant T, P , algebra gives:

Formula 4.3 Maximum Non-Expansion Work

$$-(dG)_{T,P} = \delta w_{non-PV, max}$$

Conditions

The decrease in G at constant T, P equals the maximum useful (non- PV) work extractable, e.g., electrical work from a battery.

4.4 Variation of G with pressure and temperature

Derivation. From $G = U + PV - TS$: $dG = dU + P dV + V dP - T dS - S dT$. Using first law $\delta q = dU + P dV$ and $\delta q_{rev} = T dS$:

$$dG = V dP - S dT$$

Formula 4.4 Fundamental Equation of G

$$dG = V dP - S dT$$

Derivation. At constant T ($dT = 0$): $dG = V dP$. For n moles of ideal gas, $V = nRT/P$:

$$\Delta G = \int_{P_1}^{P_2} \frac{nRT}{P} dP = nRT \ln \frac{P_2}{P_1}$$

Formula 4.5 G at Constant Temperature

$$\left(\frac{\partial G}{\partial P}\right)_T = V \quad \Delta G = nRT \ln \frac{P_2}{P_1} = 2.303 nRT \log \frac{P_2}{P_1}$$

Conditions

G increases with pressure for every substance, most steeply for gases (largest molar volume).

At constant P ($dP = 0$): $dG = -S dT$.

Formula 4.6 G at Constant Pressure

$$\left(\frac{\partial G}{\partial T}\right)_P = -S$$

Conditions

G decreases with rising T for every substance, most steeply for gases (largest entropy) – explaining why gas phase becomes relatively less stable at low T .

4.5 For a chemical reaction (pressure dependence)**Formula 4.7**

$$d(\Delta_r G) = \Delta_r V dP - \Delta_r S dT$$

At constant T , if $\Delta_r V$ is roughly constant (condensed-phase equilibria, e.g. graphite $\rightleftharpoons$ diamond, rhombic $\rightleftharpoons$ monoclinic sulfur):

Formula 4.8

$$\Delta_r G_{P_2} - \Delta_r G_{P_1} = \Delta_r V (P_2 - P_1)$$

4.6 Standard Gibbs free energy and equilibrium constant

Derivation. For a general reaction, relate the reaction Gibbs energy to the standard value and the reaction quotient Q (from the pressure-dependence result applied to each species):

$$\Delta G = \Delta G^\circ + RT \ln Q$$

At equilibrium $\Delta G = 0$ and $Q = K$:

Formula 4.9 Van't Hoff Isotherm

$$\Delta G = \Delta G^\circ + RT \ln Q \quad \Delta G^\circ = -RT \ln K = -2.303 RT \log K$$

Sign test

$K > 1 \Rightarrow \Delta G^\circ < 0$. $K = 1 \Rightarrow \Delta G^\circ = 0$. $K < 1 \Rightarrow \Delta G^\circ > 0$. Use kelvin for T ; never mix $\ln$ and $\log_{10}$ without the 2.303 factor.

Formula 4.10 Standard Gibbs Energy of Formation

$$\Delta_r G^\circ = \sum \nu \Delta_f G^\circ_{products} - \sum \nu \Delta_f G^\circ_{reactants} \quad \Delta_f G^\circ_{element} = 0$$

Key Points

Reversible phase transitions (at T_{tr} , 1 atm) always have $\Delta G = 0$, e.g. $\Delta G = 0$ for $\text{H}_2\text{O}(l) \rightarrow \text{H}_2\text{O}(g)$ at 373 K, 1 atm.

Spontaneity from sign of $\Delta H, \Delta S$

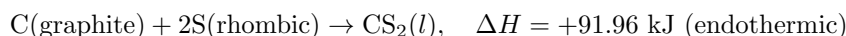
ΔH	ΔS	$\Delta G = \Delta H - T\Delta S$	Behaviour	Crossover temperature: $T_{eq} = \Delta H/\Delta S$
–	+	always –	spontaneous at all T	
–	–	– at low T	spontaneous only below some T	
+	+	– at high T	spontaneous only above some T	
+	–	always +	non-spontaneous at all T	

(where $\Delta G = 0$).

5 Thermochemistry

5.1 Exothermic / endothermic reactions

A thermochemical equation states both the material change and heat change, e.g.



5.2 Standard enthalpy of reaction

Formula 5.1 Enthalpy of Reaction

$$\Delta_r H = \sum H_{\text{products}} - \sum H_{\text{reactants}}$$

Factors affecting ΔH

Physical states of reactants/products (e.g. ΔH differs for $\text{H}_2\text{O}(\text{l})$ vs $\text{H}_2\text{O}(\text{g})$ product); quantities of reactants (extensive, scales with moles); allotropic form (diamond vs graphite); temperature and pressure (standard state = 298.15 K, 1 bar).

5.3 Hess's Law

Formula 5.2 Hess's Law of Constant Heat Summation

$$\Delta H_{\text{overall}} = \sum_i \Delta H_i \text{ (independent of path/number of steps)}$$

Key Points

Direct consequence of H being a state function. Reversing an equation reverses the sign of ΔH ; multiplying an equation by k multiplies ΔH by k . Used to combine thermochemical equations algebraically to get a target reaction's ΔH .

5.4 Standard enthalpy of formation

Formula 5.3 Enthalpy of Formation

$$\Delta_f H^\circ_{\text{element (stable form)}} = 0$$

Derivation. Since formation enthalpies are tabulated relative to elements, and $\Delta_r H = \sum H_{\text{products}} - \sum H_{\text{reactants}}$, replacing absolute H by formation enthalpies (valid because the elemental baseline cancels) gives:

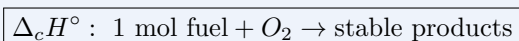
Formula 5.4 Standard Heat of Reaction from Formation Enthalpies

For $aA + bB \rightarrow cC + dD$:

$$\Delta_r H^\circ = [c \Delta_f H^\circ(C) + d \Delta_f H^\circ(D)] - [a \Delta_f H^\circ(A) + b \Delta_f H^\circ(B)]$$

5.5 Enthalpy of combustion & calorific value

Formula 5.5 Combustion & Calorific Value



$$\text{Calorific value} = \frac{|\Delta_c H|}{\text{molar mass}}$$

Derivation. Application (disproportionation via combustion data). For $\text{C}_6\text{H}_{14} \rightarrow \text{C}_4\text{H}_8 + \text{C}_2\text{H}_6$, build a Hess cycle through the common combustion products ($\text{CO}_2 + \text{H}_2\text{O}$):

Formula 5.6

$$\Delta H_{\text{reaction}} = \Delta_c H(\text{reactants}) - \Delta_c H(\text{products})$$

5.5.1 Experimental determination – bomb calorimeter

Derivation. At constant volume, heat released raises the calorimeter+water temperature by ΔT ; with heat capacity C of the system:

Formula 5.7 Constant-Volume Calorimetry

$$\Delta U = -\frac{q_{\text{comb}}}{n} = -\frac{C \cdot \Delta T}{(w_{\text{sample}}/M_{\text{sample}})}$$

Then convert to ΔH using $\Delta H = \Delta U + \Delta n_g RT$ (see §5.6). A constant-pressure bomb calorimeter (piston-fitted) gives ΔH directly by the analogous expression.

5.6 Relation between $\Delta_r H$ and $\Delta_r U$

Derivation. $\Delta_r H = \Delta_r U + \Delta(PV)$. If all reactants/products are ideal gases, $PV = n_g RT$ at fixed T , so $\Delta(PV) = \Delta n_g RT$, where $\Delta n_g = (\text{moles gaseous products}) - (\text{moles gaseous reactants})$:

Formula 5.8 ΔH – ΔU Relation

$$\Delta_r H = \Delta_r U + \Delta n_g RT$$

Conditions

Ignore liquids and solids when counting Δn_g (their molar volume is negligible vs. gases). For real gases use the full $\Delta_r H = \Delta_r U + \Delta(PV)$.

5.7 Kirchhoff's equation – variation of ΔH with temperature

Derivation. Consider $A \rightarrow B$ at T_1 and at T_2 , connected by a Hess cycle: heat A from $T_2 \rightarrow T_1$ (using $C_{p,A}$), react at T_1 , then heat B from $T_1 \rightarrow T_2$ (using $C_{p,B}$). Enthalpy being a state function around the closed cycle:

$$\Delta H_2 - \Delta H_1 = \int_{T_1}^{T_2} (C_{p,B} - C_{p,A}) dT = \int_{T_1}^{T_2} \Delta C_p dT$$

Formula 5.9 Kirchhoff's Equation

$$\Delta H(T_2) - \Delta H(T_1) = \int_{T_1}^{T_2} \Delta C_p dT$$

$$\Delta H_2 = \Delta H_1 + \Delta C_p(T_2 - T_1) \quad (\Delta C_p \text{ const})$$

Conditions

$\Delta C_p = \sum C_{p,\text{products}} - \sum C_{p,\text{reactants}}$. The analogous relation for ΔU at constant volume: $\Delta U(T_2) - \Delta U(T_1) = \int \Delta C_v dT$.

5.8 Hess's Law applications: formation, combustion, bond energy

Formula 5.10 Formation-Based Reaction Enthalpy

$$\Delta_r H^\circ = \sum \nu \Delta_f H^\circ(\text{products}) - \sum \nu \Delta_f H^\circ(\text{reactants})$$

Derivation. Bond-enthalpy estimate. Formally, break all reactant bonds to gaseous atoms (cost = $\sum E_{\text{broken}}$), then form all product bonds (release = $\sum E_{\text{formed}}$):

Formula 5.11 Bond Enthalpy Estimate of ΔH

$$\Delta H_{rxn} \approx \sum E_{broken}(\text{reactants}) - \sum E_{formed}(\text{products})$$

Conditions

For diatomic molecules, bond dissociation energy = bond energy exactly. For polyatomics, bond energy is the *mean* of the various bond-dissociation energies of that bond type (e.g. average of 4 successive C–H dissociations in CH₄).

5.9 Resonance energy from bond enthalpy

Derivation. Resonance stabilises a molecule below its single "most stable canonical (Kekulé) structure" energy. Since resonance makes bonds harder to break, the *actual* bond-breaking energy exceeds the theoretical (no-resonance) value by the resonance energy. Constructing the Hess cycle $A(g) + B^*(g) \rightarrow C(g) + D^*(g)$ through common gaseous elements (as in the source) and equating both routes gives:

Formula 5.12 Resonance Energy

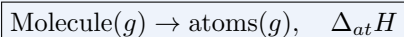
$$\Delta H_{actual} - \Delta H_{theoretical} = RE_{products} - RE_{reactants}$$

Conditions

Resonance energy is reported as a negative number (stabilisation). For benzene: compare $\Delta_f H_{actual}^\circ$ (from combustion data) against $\Delta_f H_{theoretical}^\circ$ built from Kekulé-structure bond enthalpies (3 C–C + 3 C=C + 6 C–H); the difference is the resonance energy.

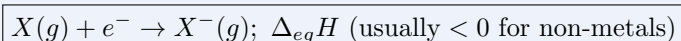
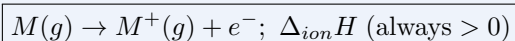
5.10 Enthalpy of atomisation

Formula 5.13



5.11 Enthalpy of ionisation & electron-gain enthalpy

Formula 5.14 Ionisation & Electron-Gain Enthalpy



Conditions

Adding a second electron to an already-negative ion is endothermic due to repulsion: e.g. $O^- + e^- \rightarrow O^{2-}$, $\Delta_{eg}H = +ve$.

5.12 Enthalpy of solution & hydration

Formula 5.15 Solution and Hydration

$$\Delta H_{sol} = \Delta H_{lattice \text{ separation}} + \Delta H_{hydration}$$

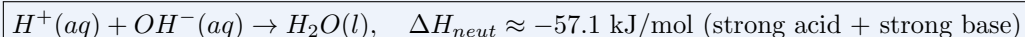
Conditions

$\Delta H_{lattice \text{ separation}} = -U$ where U is lattice *formation* energy (always endothermic to separate ions). If $|U_{lattice}| > |\Delta H_{hydration}|$, then $\Delta H_{sol} > 0$ (endothermic dissolution, e.g. KCl); if $|U_{lattice}| < |\Delta H_{hydration}|$, then $\Delta H_{sol} < 0$ (exothermic, e.g. CaCl₂). Even when $\Delta H_{sol} > 0$, dissolution can be spontaneous due to the favourable $+\Delta S$ of mixing ($\Delta G = \Delta H - T\Delta S < 0$).

5.13 Enthalpy of neutralisation

Derivation. For a strong acid + strong base, both fully ionised, so the net process is simply $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$:

Formula 5.16 Enthalpy of Neutralisation



Conditions

Defined per gram-equivalent, not per mole of acid/base – because polyprotic acids (e.g. H_2SO_4 , H_3PO_4) release multiple H^+ per mole, but each gram-equivalent still releases exactly the same -57.1 kJ for strong-strong pairs.

Derivation. Weak acid/weak base correction. Neutralising a weak acid HA with a strong base first requires ionising HA (endothermic, costs ΔH_{ion}), then the ions neutralise as usual:

Formula 5.17 Neutralisation with Weak Acid/Base

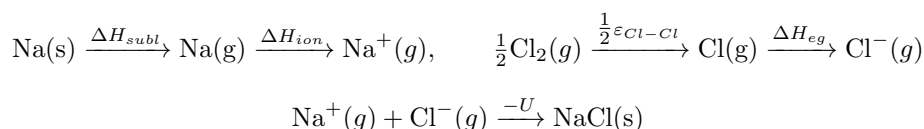
$$\Delta H_{neut}(HA/BOH) = \Delta H_{ion}(HA) + \Delta H_{ion}(BOH) + \Delta H_{neut}(H^+/OH^-)$$

Conditions

Since $\Delta H_{ion} > 0$ typically (weak electrolyte ionisation is endothermic) and $\Delta H_{neut}(H^+/OH^-) = -57.1 \text{ kJ}$, neutralisation of a weak acid/base is *less* exothermic in magnitude than $-57.1 \text{ kJ/equivalent}$.

5.14 Born–Haber cycle (lattice enthalpy)

Derivation. Lattice enthalpy U cannot be measured directly. Build a Hess cycle for e.g. $NaCl(s)$ formation from elements, passing through gaseous atoms and ions:



Since the direct route ($Na(s) + \frac{1}{2}Cl_2(g) \rightarrow NaCl(s)$) equals the sum of the indirect steps:

Formula 5.18 Born–Haber Cycle for NaCl-type (1:1) Salts

$$\Delta_f H^\circ = \Delta H_{subl} + \Delta H_{ion} + \frac{1}{2}\varepsilon_{Cl-Cl} + \Delta H_{eg} + U$$

For a salt MX_2 (e.g. $MgCl_2$), with two ionisation steps and two electron-gain steps:

Formula 5.19 Born–Haber for MX_2 -type Salts

$$\Delta_f H^\circ = \Delta H_{subl} + (I_1 + I_2) + \varepsilon_{X-X} - 2\Delta H_{eg}(X) + U$$

Conditions

All quantities are per mole of ionic compound formed; U here is defined as *released* on lattice formation, so it enters with a + sign in the enthalpy balance (equivalently $\Delta H_{lattice \text{ formation}} = -U_{lattice \text{ energy magnitude}}$, sign conventions vary by textbook – always check which is being used).

5.15 Summary table of all named enthalpy changes

Enthalpy type	Defining process	Symbol
Formation	Element(s, stable) $\rightarrow$ 1 mol compound	$\Delta_f H^\circ$
Combustion	1 mol substance + $O_2 \rightarrow$ stable oxides	$\Delta_c H^\circ$
Fusion	1 mol solid $\rightarrow$ 1 mol liquid, at T_f	$\Delta_{fus} H^\circ$
Vapourisation	1 mol liquid $\rightarrow$ 1 mol gas, at T_b	$\Delta_{vap} H^\circ$

Sublimation	1 mol solid $\rightarrow$ 1 mol gas	$\Delta_{sub}H^\circ$
Atomisation	1 mol substance $\rightarrow$ gaseous atoms	$\Delta_{at}H^\circ$
Reaction	General $A \rightarrow B$	Δ_rH°
Neutralisation	1 mol H ₂ O formed from acid+base	Δ_nH°
Ionisation	Gaseous atom $\rightarrow$ gaseous cation $+e^-$	$\Delta_{ion}H^\circ$
Electron gain	Gaseous atom $+e^- \rightarrow$ gaseous an-ion	$\Delta_{eg}H^\circ$
Lattice	Ionic solid $\rightarrow$ isolated gaseous ions	Δ_LH° or U
Bond dissociation	1 mol bonds A–B broken, gas phase	$\Delta H_{A-B} = \varepsilon_{A-B}$
Solution	Solute dissolved to infinite dilution	$\Delta_{sol}H^\circ$
Hydration	Gaseous ion $\rightarrow$ aqueous ion	$\Delta_{hyd}H^\circ$
Mixing	Pure substances $\rightarrow$ mixture	$\Delta_{mix}H^\circ$

6 Master Formula Atlas

Family	Formula	Condition / Keyword
Ideal gas	$PV = nRT$	ideal gas eq. of state
Real gas	$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$	van der Waals
First law	$\Delta U = q + w$	always (closed system)
PV work	$w = -\int P_{ext} dV$	sign: on system
Enthalpy	$H = U + PV; q_p = \Delta H$	constant pressure
Mayer	$C_P - C_V = R$	ideal gas, per mole
Equipartition	$U = \frac{f}{2}nRT$	ideal gas
Isochoric	$w = 0, q_V = \Delta U = nC_V\Delta T$	V constant
Isobaric	$w = -nR\Delta T, q_P = \Delta H = nC_P\Delta T$	P constant
Isothermal work	$w = -nRT \ln(V_2/V_1)$	T const, reversible
Isothermal ΔS	$\Delta S = nR \ln(V_2/V_1)$	T const
Adiabatic (PV)	$PV^\gamma = \text{const}$	$q = 0$, reversible
Adiabatic (TV)	$TV^{\gamma-1} = \text{const}$	same
Adiabatic (TP)	$T^\gamma P^{1-\gamma} = \text{const}$	same
Adiabatic work	$w = \frac{nR(T_2 - T_1)}{\gamma - 1}$	$q = 0$
Polytropic	$PV^n = \text{const}; w = \frac{P_2V_2 - P_1V_1}{n - 1}$	general path index n
Entropy (def)	$dS = \delta q_{rev}/T$	reversible heat
Ideal-gas entropy	$\Delta S = nC_V \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1}$	ideal gas
Carnot efficiency	$\eta = 1 - T_C/T_H$	reversible engine
Clausius inequality	$\oint \delta q/T \leq 0$	any cyclic process
Total entropy	$\Delta S_{sys} + \Delta S_{surr} \geq 0$	spontaneity criterion
Gibbs function	$G = H - TS$	definition
G differential	$dG = V dP - S dT$	pure substance
G vs T	$\Delta G = \Delta H - T\Delta S$	constant T
G criterion	$\Delta G < 0$ spontaneous	constant T, P
G isothermal (gas)	$\Delta G = nRT \ln(P_2/P_1)$	ideal gas, const. T
Van't Hoff	$\Delta G = \Delta G^\circ + RT \ln Q$	general
isotherm		
Equilibrium	$\Delta G^\circ = -RT \ln K$	at equilibrium
Formation ($\Delta_r H$)	$\Delta_r H^\circ = \sum \nu \Delta_f H_{prod}^\circ - \sum \nu \Delta_f H_{react}^\circ$	Hess's law form
Kirchhoff	$\Delta H_2 - \Delta H_1 = \int_{T_1}^{T_2} \Delta C_p dT$	temperature correction
$\Delta H - \Delta U$	$\Delta_r H = \Delta_r U + \Delta n_g RT$	gas-phase reaction
Bond estimate	$\Delta H \approx \sum E_{broken} - \sum E_{formed}$	bond energy data
Resonance energy	$\Delta H_{actual} - \Delta H_{theor} = RE_{prod} - RE_{react}$	bond-energy method
Born-Haber (1:1)	$\Delta_f H^\circ = \Delta H_{subl} + \Delta H_{ion} + \frac{1}{2}\varepsilon + \Delta H_{eg} + U$	ionic solid formation
Hess's law	$\Delta H_{overall} = \sum_i \Delta H_i$	path independence
Third law	$S(T=0) = 0$	perfect crystal

7 Keyword Decision Tree

Recognise the process from the question

- “**T constant**” / “isothermal” → isothermal formulas ($w = -nRT \ln(V_2/V_1)$, $\Delta U = 0$)
- “**no heat exchange**” / “insulated” / “ $q = 0$ ” + reversible → adiabatic ($PV^\gamma = \text{const}$)
- “ $PV^n = \text{constant}$ ”, general index n → polytropic
- “**constant volume**” / rigid vessel → $w = 0$, $q_V = \Delta U$
- “**constant pressure**” / open vessel → $q_P = \Delta H$
- “**cycle**” with source & sink → Carnot efficiency
- “**entropy**”, “disorder” → ΔS relations (ideal gas / phase transition / reaction)
- “**spontaneous**” at given T, P → ΔG criterion
- “**temperature correction**” to ΔH → Kirchhoff’s equation
- “**combine reactions**”, given several ΔH ’s → Hess’s law
- “**lattice energy**” of an ionic solid → Born–Haber cycle
- “**bond energies given**”, estimate ΔH → bond enthalpy method
- “**K given**”, find ΔG° (or vice versa) → $\Delta G^\circ = -RT \ln K$

8 Symbol & Sign Glossary

Quick reference

U = internal energy (state fn.) $H = U + PV$ (state fn.) S = entropy (state fn.) $G = H - TS$ (state fn.)
 q, w = path functions.

$w > 0$: work done *on* system (compression). $w < 0$: work done *by* system (expansion).

$q > 0$: heat absorbed by system. $q < 0$: heat released by system.

$\Delta H < 0$, $\Delta U < 0$: exothermic. $\Delta H > 0$, $\Delta U > 0$: endothermic.

$\gamma = C_P/C_V > 1$ always, for any gas. Δn_g = moles gaseous product – moles gaseous reactant (solids/liquids excluded).

Always use **kelvin** for T in every ratio/log/entropy formula; use 2.303 to convert $\ln \rightarrow \log_{10}$.

Last-minute exam traps

- Kelvin, not Celsius, in every thermodynamic ratio.
- $\ln$ is natural log; $\log_{10}$ needs the factor 2.303.
- Under the convention here, gas expansion gives $w < 0$ (work done *by* the system reduces U).
- Count *only gaseous* species when computing Δn_g for $\Delta H = \Delta U + \Delta n_g RT$.
- Do not include vibrational degrees of freedom unless the question explicitly says so.
- $PV^\gamma = \text{const}$ applies to *reversible* adiabatic changes only – never to a sudden (irreversible) adiabatic change.
- Enthalpy of neutralisation is defined per gram-equivalent, not per mole of polyprotic acid/base.

End of formula & derivation book – every boxed result above is traceable to a first-principles derivation from the first law, the definition of entropy, or the definition of Gibbs free energy.